INFLAMMATION & IMMUNE DEFENSE

LL-37 — the full review

CATHELICIDIN ANTIMICROBIAL PEPTIDE

This is the deeper look behind the short version — where LL-37 comes from in the body, why researchers are now studying ways to block it rather than add more of it in some skin conditions, what the wound-healing trials actually found, and what your consultation covers.

What it is

LL-37 is the only cathelicidin-family antimicrobial peptide the human body makes. It starts out as a larger precursor protein, packaged inside immune cells called neutrophils and also produced by skin cells and the lining of the airway, gut, and urinary tract. When needed, an enzyme clips off the active 37-amino-acid piece — LL-37 — as part of the body's fast-acting, first-line immune defense.

What is sold today under the name LL-37 is a synthetic replica of that natural peptide, offered as an injectable vial or, less often, a compounded topical cream, marketed for wound healing, immune support, and general skin health. None of this is FDA-reviewed, and the skin-condition marketing in particular runs against a substantial body of research pointing the other way.

What people use it for

USED FOR	EVIDENCE	WHAT THAT MEANS
Wound healing (venous leg ulcers, diabetic foot ulcers)	**Human trials**	The only use with real controlled human data; the largest trial missed its main endpoint, with benefit limited to a smaller subgroup.
Antimicrobial and immune defense support	**Theoretical**	Well demonstrated in test-tube studies; no human trial shows it reduces infection risk or treats an active infection.
Skin conditions — psoriasis, rosacea, atopic dermatitis	**Not studied for this use**	The research literature links elevated, not deficient, LL-37 to these conditions; researchers are studying ways to block it, not add more.
General immune support or wellness	**Theoretical**	No human trial supports this framing; rests on extrapolating from LL-37's normal immune role.

How it works

LL-37's antimicrobial action comes from its structure: a positively charged shape that lets it insert into and destabilize the membranes of bacteria, fungi, and some viruses, causing them to break down. It is also chemotactic — it recruits immune cells to a site of infection or injury and promotes new blood vessel growth and skin-cell migration, the proposed basis for a wound-healing effect.

A separate, less-marketed mechanism matters just as much: LL-37 binds fragments of the body's own DNA and RNA released from damaged cells, and those complexes can trigger a strong inflammatory immune response. This is the same pathway implicated in driving, not resolving, several inflammatory and autoimmune skin conditions — a mechanism worth understanding before treating LL-37 as a simple immune booster.

What the research actually looked at

The body's own, naturally produced LL-37 is very well studied — a large, decades-deep literature on its role in immune defense, wound biology, psoriasis, rosacea, atopic dermatitis, lupus, and other conditions. That is legitimate science, but it is research on a peptide the body already regulates tightly, not evidence that giving someone additional, externally administered LL-37 is beneficial.

Administered synthetic LL-37 in humans has been tested in a small, narrow set of controlled trials — essentially four, all in wound care, totaling roughly 200 people. The largest and most rigorous of these, a 149-person trial of topical LL-37 for venous leg ulcers, did not meet its primary healing endpoint in the overall study population; a benefit only appeared in a smaller, post-hoc subgroup of larger wounds. Two smaller trials showed modest, inconsistent positive signals. No trial has tested LL-37 for immune support, antimicrobial defense, or any skin condition beyond wound care.

Common misunderstandings

- **“It's natural, so my body already makes it, and more must be safe.”** The clearest human disease evidence shows elevated, dysregulated LL-37 driving inflammation in psoriasis, rosacea, and lupus — closer to a case of too much than too little.
- **“It helps psoriasis and rosacea.”** The evidence points the other way: both conditions show abnormally high LL-37 activity in affected skin, and researchers are pursuing ways to block its signaling as a treatment direction.
- **“There's strong clinical proof it heals wounds.”** The largest, most rigorous trial did not meet its primary healing endpoint in the overall study population; the positive signal that exists is limited to a smaller subgroup.
- **“It's a cancer-fighting immune peptide.”** The research literature describes genuinely mixed, tissue-specific effects — appearing to promote tumor growth in some cancer types while suppressing it in others.

Who it is not for

- Anyone with an autoimmune or autoinflammatory condition, including psoriasis, lupus, or rosacea.
- Anyone with a personal or strong family history of cancer, or an undiagnosed mass.
- Pregnant or breastfeeding patients, or anyone trying to conceive.
- Anyone with an active infection who needs established antimicrobial treatment rather than an unproven alternative.
- Anyone sourcing product from an unverified research-use-only vendor.
- Competitive athletes subject to WADA testing.

Competitive athletes: LL-37 is not currently named specifically on the WADA Prohibited List, but WADA's S0 category ("Non-Approved Substances") prohibits, at all times, any substance without current approval by a governmental health authority for human therapeutic use — a category that has been applied to other unapproved peptides. Athletes in tested sport should confirm current status with their sport's anti-doping authority before use.

Who makes this, and under what rules

Everything dispensed here is prepared by a licensed 503A compounding pharmacy — not bought from a supplier and repackaged, and not sourced from an online seller.

That designation carries specific, enforceable obligations:

- **A prescription in your name.** Compounded from a prescription for one identified patient, after an evaluation.
- **Licensed and inspected.** A state licence, subject to inspection by the state board of pharmacy.
- **Ingredients that can be traced.** A certificate of analysis from an FDA-registered facility, back to a named source and lot.
- **Sterile preparation to a published standard.** USP standards.
- **A paper trail.** A lot number and beyond-use date on every preparation.

A compounded preparation is not an FDA-approved medicine — but it is the difference between a licensed pharmacy working to a standard and an unregulated website.

Where this stands

Never approved, and its compounding status is unsettled. Not on FDA's active compounding lists — it sits in a withdrawn-nomination limbo, and a future FDA advisory review is reportedly expected before early 2027.

The human trial record is small, and its flagship trial missed its goal. Roughly 200 people studied across four small wound-care trials; the largest, most rigorous one did not meet its primary endpoint in the overall population.

For skin conditions specifically, the research points toward blocking LL-37, not adding more of it.

Clinical detail

The only human cathelicidin, cleaved from precursor protein hCAP-18 by proteinase 3 in neutrophils; also expressed by skin and epithelial cells.

Sits in FDA's "nominated but withdrawn" table for 503A bulk compounding substances; not on the active Category 2 significant-safety-risk list either.

Industry trackers report FDA-identified concerns from an earlier nomination review: limited human safety data, possible immunogenicity, reproductive findings in nonclinical research, and a tumor-promotion signal — not independently confirmed against a primary FDA document in this research pass.

The largest controlled human trial (topical, venous leg ulcers, n=149) did not meet its primary efficacy endpoint in the full study population.

Peer-reviewed research links elevated (not deficient) LL-37 to the pathology of psoriasis and rosacea, and describes tissue-specific pro- and anti-tumor effects across different cancer types.

> The information on this page is general education. It is not medical advice and it is not a recommendation for any treatment.
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> LL-37 is not an FDA-approved medicine for the uses described here. A compounded preparation is not an FDA-approved product and has not been reviewed by the FDA for safety, effectiveness or quality.
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> Descriptions of what this is used for reflect how it is commonly used and discussed. They are not claims that it is effective for any purpose.
>
> Any prescription follows an individual evaluation by a licensed provider and is issued only where that provider judges it clinically appropriate. A prescription is never guaranteed.